动态随机一般均衡模型：理论与应用
第 5 章：价格黏性

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A 动态系统

A.1 Calvo 定价下经济动态系统

$$\beta \mathbb{E}_t \frac{C_{t+1}^{-\sigma}}{C_t^{-\sigma}} \frac{1}{\Pi_{t+1}} = \frac{1}{R_t}$$
$$N_t^{\varphi} C_t^{\sigma} = \hat{W}_t$$
$$Y_t = A_t N_t$$
$$\hat{\dot{P}}_t = \frac{1}{\mu} \frac{S_{1t}}{S_{2t}}$$
$$S_{1t} = C_t^{1-\sigma} \hat{MC}_t + \beta\theta \mathbb{E}_t \Pi_{t+1}^{\epsilon} S_{1t+1}$$
$$S_{2t} = C_t^{1-\sigma} + \beta\theta \mathbb{E}_t \Pi_{t+1}^{\epsilon-1} S_{2t+1}$$
$$\theta \Pi_t^{\epsilon-1} + (1-\theta)\big(\hat{\dot{P}}_t\big)^{1-\epsilon} = 1$$
$$\hat{MC}_t = \hat{W}_t / A_t$$
$$\frac{R_t}{R} = \Pi_t^{\phi_\pi}$$
$$Y_t = C_t V_t$$
$$V_t = (1-\theta)\hat{\dot{P}}_t^{-\epsilon} + \theta \Pi_t^{\epsilon} V_{t-1}$$
$$a_t = \rho a_{t-1} + \eta_t^a$$

当观察到冲击的厂商重新设定产品价格时，未来的价格并非保持不变，而是根据前一期的通胀指数化调整时，模型动态系统变为：

$$\beta \mathbb{E}_t \frac{C_{t+1}^{-\sigma}}{C_t^{-\sigma}} \frac{1}{\Pi_{t+1}} = \frac{1}{R_t}$$
$$N_t^{\varphi} C_t^{\sigma} = \hat{W}_t$$
$$Y_t = A_t N_t$$
$$\hat{\dot{P}}_t = \frac{1}{\mu} \frac{S_{1t}}{S_{2t}}$$
$$S_{1t} = S_{1t} = C_t^{1-\sigma} \hat{MC}_t + \beta\theta \mathbb{E}_t \big(\frac{\Pi_{t+1}}{\Pi_t^{\gamma_p}}\big)^{\epsilon} S_{1t+1}$$
$$S_{2t} = C_t^{1-\sigma} + \beta\theta \mathbb{E}_t \big(\frac{\Pi_{t+1}}{\Pi_t^{\gamma_p}}\big)^{\epsilon-1} S_{2t+1}$$
$$\theta \big(\frac{\Pi_t}{\Pi_{t-1}^{\gamma_p}}\big)^{\epsilon-1} + (1-\theta)\big(\hat{\dot{P}}_t\big)^{1-\epsilon} = 1$$

$$\hat{MC}_t = \hat{W}_t / A_t$$
$$\frac{R_t}{R} = \Pi_t^{\phi_\pi}$$
$$Y_t = C_t V_t$$
$$V_t = (1-\theta)\hat{\dot{P}}_t^{-\epsilon} + \theta\big(\frac{\Pi_t}{\Pi_{t-1}^{\gamma_p}}\big)^{\epsilon} V_{t-1}$$
$$a_t = \rho a_{t-1} + \eta_t^a$$

A.2 Rotemberg 定价下经济动态系统

$$\beta \mathbb{E}_t \frac{C_{t+1}^{-\sigma}}{C_t^{-\sigma}} \frac{1}{\Pi_{t+1}} = \frac{1}{R_t}$$
$$N_t^{\varphi} C_t^{\sigma} = \hat{W}_t \quad Y_t = A_t N_t$$
$$\hat{MC}_t = \hat{W}_t / A_t$$
$$\hat{MC}_t = \mu + \frac{\theta_p}{\epsilon}\left[(\Pi_t - 1)\Pi_t - \mathbb{E}_t Q_{t,t+1}(\Pi_{t+1}-1)\Pi_{t+1}^2 \frac{C_{t+1}}{C_t}\right]$$
$$\frac{R_t}{R} = \Pi_t^{\phi_\pi}$$
$$Y_t = C_t + \frac{\theta_p}{2}(\Pi_t - 1)^2 C_t$$
$$a_t = \rho a_{t-1} + \eta_t^a$$

B 求解模型稳态

设稳态时 $\Pi = 1$，那么，

$$\hat{R} = 1/\beta$$
$$\hat{W} = \mu$$
$$Y = N = C = \mu^{\frac{1}{\sigma+\varphi}}$$

C 对数线性化

C.1 无指数化调整

Calvo 定价和 Rotemberg 定价下线性系统是类似的，唯一的区别在通胀动态议程中实际边际成本的系数 κ 与结构参数的关系。

$$c_t = \mathbb{E}_t c_{t+1} - \frac{1}{\sigma}(r_t - \mathbb{E}_t \pi_{t+1})$$
$$\varphi n_t + \sigma c_t = \hat{w}_t$$
$$y_t = a_t + n_t$$
$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa \hat{mc}_t$$
$$\hat{mc}_t = \hat{w}_t - a_t$$
$$r_t = \phi_\pi \pi_t$$

$$y_t = c_t$$
$$a_t = \rho a_{t-1} + \eta_t^a$$

在 Calvo 定价下，

$$\kappa = \frac{(1-\theta)(1-\beta\theta)}{\theta}$$

在 Rotemberg 定价下，

$$\kappa = \frac{\epsilon - 1}{\theta_p}$$

C.2 有指数化调整

引入指数化调整之后，线性动态系统为：

$$c_t = \mathbb{E}_t c_{t+1} - \frac{1}{\sigma}(r_t - \mathbb{E}_t \pi_{t+1})$$
$$\varphi n_t + \sigma c_t = \hat{w}_t$$
$$y_t = a_t + n_t$$
$$(1 + \beta\theta\gamma_p)\pi_t = \kappa \hat{mc}_t + \beta \mathbb{E}_t \pi_{t+1} + \gamma_p \pi_{t-1}$$
$$\hat{mc}_t = \hat{w}_t - a_t$$
$$r_t = \phi_\pi \pi_t$$
$$y_t = c_t$$
$$a_t = \rho a_{t-1} + \eta_t^a$$